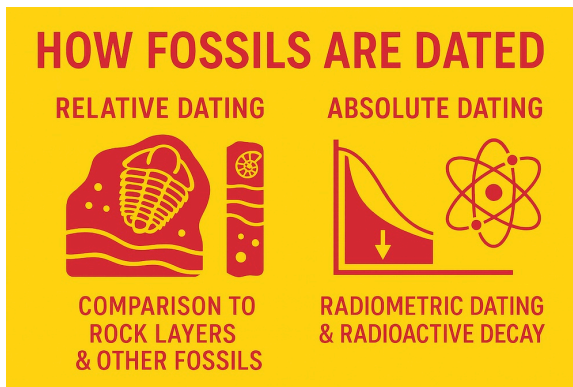


Week 7: Understanding Time

Establishing a Temporal Context

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Key Points

- **Additional Assigned Reading:** Ch. 7 in textbook; Section: Dating Methods
- **Key Terms:** Beringia, Cordilleran Ice Sheet, Laurentide Ice Sheet, Beringia Standstill, Kelp Highway, Ice-Free Corridor, Clovis/Clovis-First
- **Summary:** The earliest evidence of the peopling of the Americas sits between 18-15 ka. The ancestors of the first settlers of the Americas trace back to the Ancient Paleo-Siberians who settled in Northeast Eurasia after migrating through the Inner Asian Mountain Corridor (IAMC). While it is clear that the route into North America spans across the Bering Land Bridge, there is still much research into the path into the heart of the continent (or into S. America). The primary hypotheses which aim to explain this process are the “Kelp Road” and “Ice-Free Corridor” hypotheses.

1 Unit 2: Intent and Strategy

This unit is designed to *walk* students through the spread of our species at the tail end of the Paleolithic. The emergence of *Homo sapiens* at approximately 300 ka illustrates the last branch on our evolutionary tree but in no way means that we stopped evolving. Along with other, co-existing hominins of the time, we continued to adapt to everchanging landscapes, develop new technologies and survival strategies, and explore the great expanses of this planet that humans had never seen.

We discussed what a species *is* for some length in the last unit. While we don’t need to open the conversation of murky taxonomy back up here, I’d offer one more approach to considering what this taxon represents. Recall the term **Ontogeny** from our lecture on “Primate Characteristics”. This term refers to the developmental history of an organism throughout it’s individual lifetime. The word has also been co-opted in species discussions to reflect the evolutionary nature of a single species’ existence. We can thus consider the Ontogeny of *Homo sapiens* and in doing so come to realize that we have only discussed its infant years so far. The period of time from the origin of our species (~300 ka) until it began to *successfully* spread across the globe could, in this sense, be compared to the moment of a child’s birth until the moment it can competently walk on its own. While not a perfect analogy it should evoke some degree of understanding about what’s to come in our discussions.



Before we get to any of the material on new cultural adaptations and global exploration, however, we have to take some time to understand how it is that we know when and under what conditions our species began to navigate the Earth. That's right, buckle your seatbelts, it is time for us to have "the talk". Over the next couple of weeks we will thus be learning about *how* we "know" what we "know". Being that much of our understanding of human history and adaptation is dependant on time and environment, it is important that you come to understand our methods in reconstructing these contextual aspects of human paleontology and archaeology. I can't stress enough how important these methods are to our fields of study. Perhaps surprisingly, the methods that have become so important to us are not Anthropological methods at all but those of Geologists (Geochronology and Geochemistry/Biogeochemistry). Biological Anthropology and Archaeology are fields which are inherently multidisciplinary. In this case, we want to study human evolution and cultural development and to do so we need to use the methods employed

by Geologists to develop a **Temporal Context** and a **Paleoenvironmental Context**. So here we go...

:arrow_up: Anthropologists are Zombies! I once had a professor who referred to Anthropologists as Zombies because we have a tendency of moving across disciplines and consuming their ideas and methods. I'm not sure this was meant as an entirely positive metaphor but I kinda dig it. It's not to say that we don't develop our own ideas (we certainly do) but more to say that we do not shut ourselves off from the incredibly useful work of other disciplines. Shout out to Dr. Saetra here!

2 Dating

! Important

This information reviews and expands on the sections in your assigned reading (Ch. 7: Dating Methods).

The first thing that any student learns when it comes to establishing a temporal context in Paleontology and Archaeology is that there are two general approaches to this goal: **Relative Dating** and **Absolute (or Radiometric) Dating**. Before getting into the details of each we can summarize it with a colloquial analogy. In this hypothetical, let us say that the last time you travelled out of state was in the month of May. If someone were to ask "when was the last time you travelled out of state?" you could respond in two ways which could satisfy the question: you could say that you travelled just before the Summer or you could provide the exact date (say, 05/20/2026). The first response is less exact but in placing it in relative position to a known time/event you provide the context that is typically appropriate for the question. This is an example of a Relative Date. If the question was a bit more formal, however, this might not satisfy and you would want to provide the Absolute (or precise) Date.

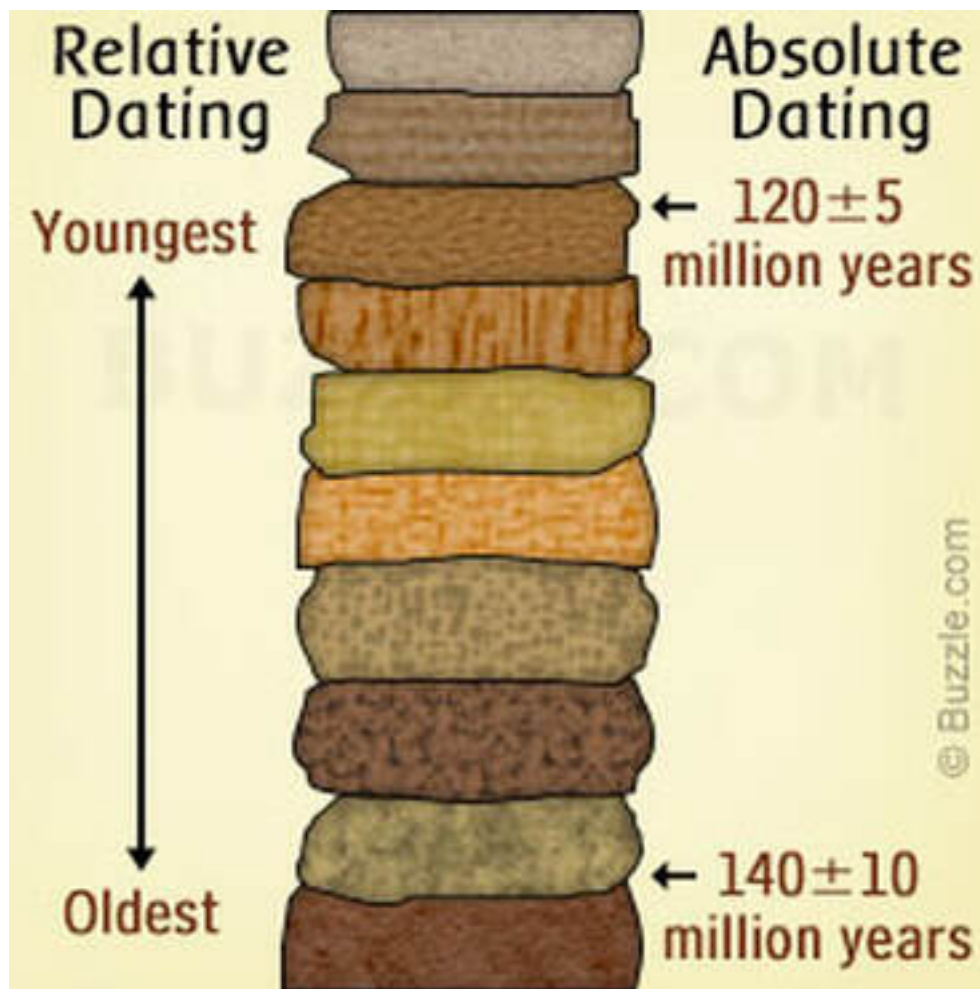


Figure 1: Two approaches to dating archaeological and paleontological sites

The image to the right highlights this. It shows successive layers of rock that we call **Strata** (see below). On the right side of these rocks are indicators of Absolute age in Millions-of-Years (which we have been abbreviating Ma). The ($\pm$) symbol tells you that there is an error range for this date. It is known that our numerical dates are seldom 100% precise. Instead they are more likely to provide a range of dates within a certain degree of confidence. Here the upper date is really suggesting that that stratum dates between approximately 115-125 Ma. More on that later. On the left of the rocks, under Relative Dating, there is simply an arrow showing that older rocks are on the bottom of this pile and younger ones are on top. This observation, known as **The Law of Superposition**, is really only part of the story. As suggested in my analogy above, to make this a bit more explicit we would indicate that one of these layers represents a specific event (say like the date that you traveled out of state or, more to the point, the extinction of dinosaurs). With that we could use the relative dating scheme to say that some strata are older than the extinction of dinosaurs and some are younger (like saying that you travelled before or after the Summer). While this brief overview should hopefully get the general difference between the two across, there are some important details of each that you will need to better understand in this class.

3 Relative Dating

The beauty of relative dating is in its ability to construct a logical timeline. In general discussion of history, specific dates are details that are only necessary with narrow inquiry. If we reflect on our timeline of human evolution, for example, I'd be happy if you could recount that *Ardipithecus* came before *Australopithecus* which came before *Homo*. This is a relative dating timeline that can be redrawn in a couple of ways [Diagram 1].

3.0.a A



3.0.b B

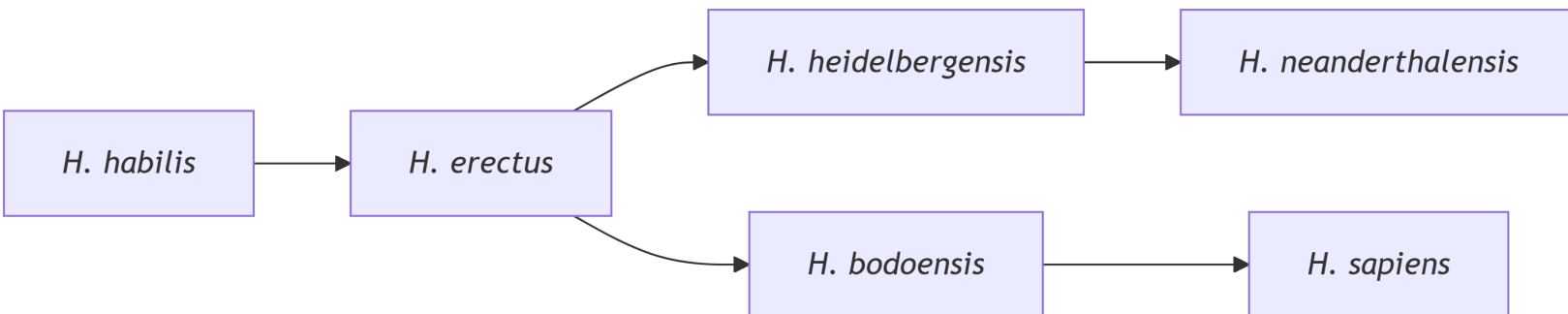


Figure 2: Both timelines illustrate the logic of relative dating. Though they do not give you numerical ages, such timelines do well to illustrate an order of events. In many discussions of history, these illustrations are sufficient to convey primary points. For instance, it was important for us to know that bipedalism evolved before increased encephalization. I should also point out that I used two controversial designations in the bottom timeline: *H. heidelbergensis* and *H. bodoensis*. For more information on this layout refer back to our discussion in week 5 on “Post-erectus hominins”.

3.0.c Sedimentary Rocks

The foundation of relative dating methods is situated in Sedimentology (the study of sedimentary rocks and the processes by which they are formed). If by chance you read the non-assigned sections of Chapter 7 in the textbook, you have learned that Sedimentary, Igneous, and Metamorphic are the three primary rock classes in Geology. We are interested in Sedimentary rocks here because (1) they are most likely to preserve fossils and (2) because they form in predictable chronological ways. As suggested by the name, Sedimentary rocks are formed by sediments of other materials. While largely sediments from other rocks, they are also regularly comprised of other materials (organic or inorganic). The original material (the sediments) come from the process of **erosion** which all matter is subjected to. Be it wind, rain, sun, or simply time, all things erode into smaller pieces. When this happens, the sediments begin a new life detached from their parent structure.

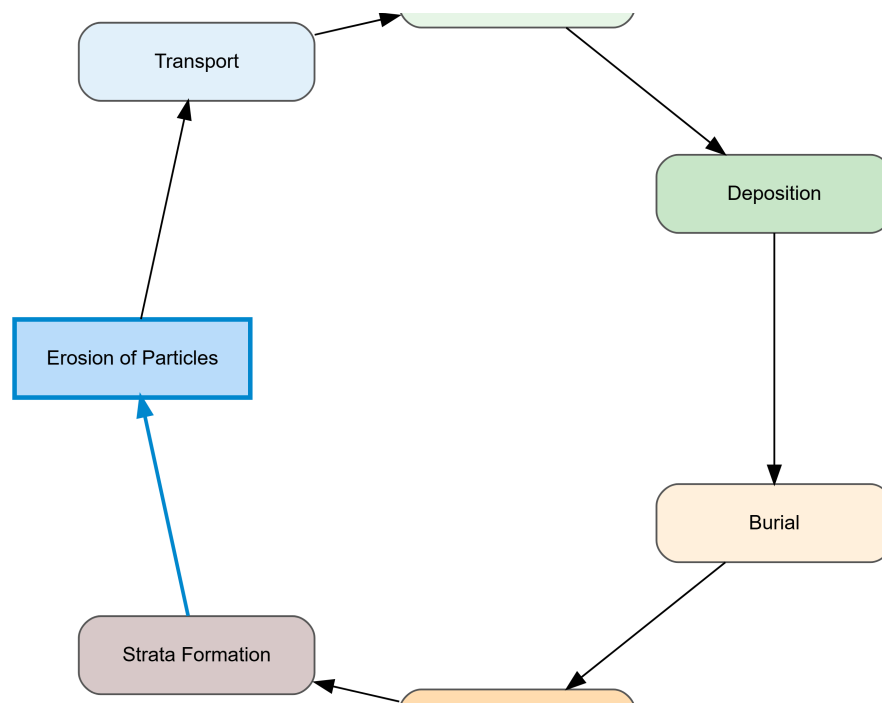
Depending on the size of the eroded particle, a variety of energetic forces now act to move it from one spot to another. Energetic disparity across different regions therefore creates areas of high erosion (high energy environments) and areas of **deposition** (lower energy environments where sediments are deposited). Depositional environments are where we see the formation of sedimentary rocks and, ultimately, the best preservation of fossils.

3.0.d Strata, Stratigraphy, and Superposition

To understand Relative Dating you must have knowledge of some rudimentary terms. Both **Stratigraphy** and the **Law of Superposition** are mentioned in the text, so we don't need to spend too much time on them here. Stratigraphy is simply the study of **Strata**. While the text doesn't provide much of a definition for strata, the term also quickly becomes self evident: each identifiable layer of sedimentary rock is referred to as a single stratum (*singular*) or, when multiple are discussed, layers of Strata (*plural*). With these details out of the way we can start to envision what is really meant by **The Law of Superposition**. Sediments are carried from the original structure that they eroded from by some energetic force (gravity, wind, water...) until the energy dwindles enough for the weight of the particle to overcome the transporting energy. Then it falls (becoming sediment). Being a low energy environment, this particle is not the only one that falls of in the area. Anything that has been energetically transported here faces this fate and becomes deposited as sediment just the same. We therefore have a "bed" of sediments that forms. This is essentially a sandbox (or a lake bed, river bed, and etcetera).

[Sidenote: understanding the energy requirements for sediment transportation is actually pretty cool. When looking at depositional environments you can begin to form an idea of the original setting. If, for instance, you are hiking and come across a dried up river bed that has large stones scattered throughout, you can predict that the river was flowing fast and hard enough at this point to carry such large stones. If you then continue to follow it and notice that the stones get smaller, or just become beds of sand, you can observe how the flow of water changed in different parts of the river. Maybe it's not all that cool to you, but I love this stuff. Such understanding has a tendency of making the world seem infinitely more awesome.]

OK, now picture this depositional environment. It could be a sand dune or the bottom of a lake or ocean. In any case sediments are slowly accumulating and burying whatever lied there before. As more and more sediment piles up, the underlying material becomes subjected to more and more pressure and heat. Combine this with the groundwater that percolates through these materials and the sediments begin to literally glue together. This is the process of Sedimentary Rock formation. From here we can have little doubt what the **Law of Superposition** is all about. Physical structures succumb to erosion, eroded particles are carried away from their parent structures until the transporting energy dissipates and they become sediments, sediments accumulate in depositional environments and bury the sediments that came before them, and finally the buried sediments (under the forces of pressure, heat, and moisture percolation) glue together to form distinct layers of rock. These are strata. The deepest strata, by necessity, formed before the strata above them. BAM! Superposition. Suffice it to say that this cycle starts all over again when that stratum is eventually exposed again to the elements and begins to erode. Check out the diagram below for an overview of this process [*Diagram 2*].



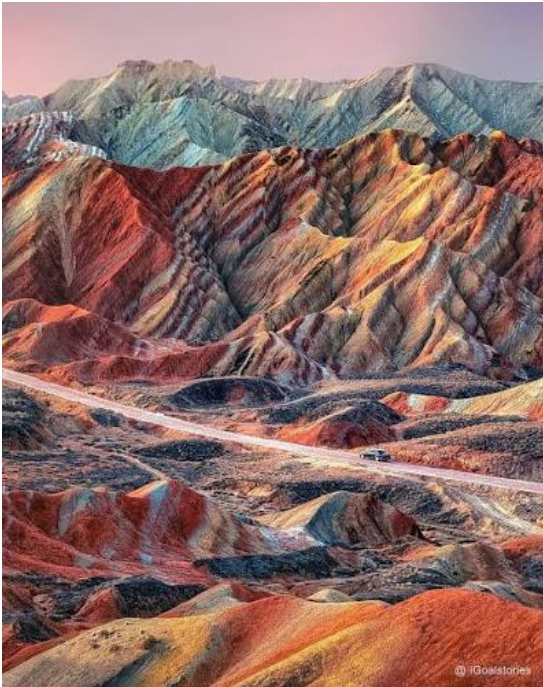
3.1 The Sedimentary Rock Cycle

The processes of sedimentation, cementation, and erosion are commonly considered to be Uniformitarian processes. It is a constant process by which our planet continues to recycle and reshape its principle structure.

Figure 3: Lifecycle of Sedimentary Rocks

But wait...how do we tell different strata apart? What makes them look different and what does this have to do with fossils and time?

One of the beautiful things about sedimentary rocks being used as time markers is that they can often tell you about shifts in the environment. Darker strata, for instance, commonly reflect regions abundant with life where the decomposition of organic matter mixes with sediments to make these darker hues. White horizons could signify an abundance of quartz or could be dense bands of limestone (made up of shells from microscopic marine organisms). Bands of red strata might suggest that this layer was rich in iron which subsequently oxidized when exposed to high oxygen environments just as they can turn green in oxygen reducing environments. Whatever the case, the sedimentary rocks preserve the conditions of the depositional environments from which they formed. They lock these evidentiary materials in stone and store them in the geological record.



(a) An image from Zhangye National Park in Gansu, China. Known to be among the most colorful geological sequences in the world, this exposed landscape is comprised of sandstone and siltstone rich in iron oxide. Minerals such as hematite, chlorite, and goethite are illustrated in the colorful stratigraphic bands



(b) The Painted Hills are a geological unit within the John Day Fossil Beds in Mitchell, Oregon. In addition to the iron oxide noticeable from the red bands, the Painted Hills boast layers of Volcanic Rhyolite (igneous material) that fell over the landscape and created horizons rich in deep purples.

Figure 4: Examples of Stratigraphy from around the world

3.1.a Fossils and Time

From here there should be little problem understanding how the relative dating of fossils takes place. These strata are far from isolated. In fact, many **Stratigraphic Horizons** represent the same period of time across huge continental masses. But it is not just the sedimentary matter that helps us to identify and define them. Even more useful to this cause are the fossils that get buried within these horizons in successive layers. Whether they are transported by the same energetic forces as the sediments themselves or they simply become buried in the environments that they die in, plants, animals, and even bacteria can become preserved in these strata. Such environments create tombs which encase the most structurally sound components of the organisms that get buried within (often shells, bones, and teeth). What this means is that successive strata have the ability to preserve a record of the life that existed on our planet hundreds, thousands, millions, and even billions of years ago. This is the record that allows to definitively say, “No, Dinosaurs and Humans did not coexist. Dinosaurs went extinct long before the first human-like animals ever existed. In fact, barring the continued success of archosaurs (including dinosaurs but currently represented by crocodilians and birds), dinosaurs went extinct long before even the first primates

evolved.” This is a succinct and scientifically accurate relative dating account of the history of life on our planet.

But wait, there’s more...

The most precise way to apply relative dating is in combination with absolute dating. Your assigned reading makes mention of **Biostratigraphy** but really does not give it its due justice. It states that “this form of dating looks at the context of a fossil or artifact and compares it to the other fossils and biological remains (plant and animal) found in the same stratigraphic layers.” I just don’t think that tells you much, so here is my attempt. First, one of the founding principles of biostratigraphy is that it can match strata from different parts of the world. Let’s say you have two stratigraphic sequences in different regions that look the same but vary in depth. This is a common occurrence. The strata might be color patterned (say grey, green, black, grey, white, red) in East Africa and India. But in India the red stratum is two meters deep and in East Africa it is barely visible (or absent altogether). You might have a thought that they represent the same period of time but you really can’t tell. The best possible way of solving this riddle is to look at the fossils. Having not discussed it much, it may seem like the inclusion of fossils in strata is a rarity. Not really. There is a lot of life on this planet and most of it gets buried after death. While fossilization is a relatively rare occurrence, even a rare occurrence can have a large yield if the base material is abundant. In the right environment, fossils should be an expectation not a surprise. In most stories of fossil hunting (including my own) the struggle is not finding fossils *per se* but finding the desired fossils. In my previous research, for example, I was part of a team investigating the presence of *H. erectus* in a specific region of Ethiopia. While this was a daunting task which provided only a few quality specimens from this species, the region provided thousands of fossils from other animals. Though not our primary research goal, one important aspect of this project was establishing the specific context of the region at the time that *H. erectus* lived in it. It is largely due to these fossils and their quality as Time Markers and Environmental Indicators that we were successful in this goal. We will come back to the notion of fossils being environmental indicators in the next lecture. For now, let us focus on the massive potential of **Biostratigraphy** and its use in Relative Dating.

3.1.b Time Markers



Figure 5: Redlichiida trilobite indicative of an Early to Middle Cambrian age.

Biostratigraphy is the study of fossil remains in stratigraphic contexts in order to (1) correlate strata from different environments and (2) provide evidence for the geologic age of strata within a stratigraphic sequence (see Table 1). The first aspect of this, referred to more specifically as **Fuunal Correlation**, is discussed above. This process allows geologists and paleontologists to match strata from different parts of the world which may have been separated through other geologic/geographic actions (plate tectonics, volcanic eruption, erosion of river valleys, and et cetera). The second approach to Biostratigraphy is the explicit use of fossils as time markers. Key to this effort is the identification of **Index Fossils**.

Applications of Biostratigraphy

Table 1: Applications of Biostratigraphy

Biostratigraphy	Description
Correlate strata	Matching strata from different regions provides context for a study area. Even if you don't know the age, the correlated strata can add ecological information by identifying fossil species present in one sample but not the other, it can inform on geologic/geographic processes such as plate tectonics and volcanic activity, and establish a relative timeline across the fossil record.
Identify geologic age	While similar to faunal correlation, identifying a geologic age is a specific agenda which uses distinct fossils sets and known numeric ages to provide a narrow age range for particular stratigraphic units.

Index Fossils come from species which are abundant in the fossil record, easy to identify, and have a relatively short geologic age. In meeting these criteria, these fossils not only have the power to identify specific strata from different regions (faunal correlation), but to delineate the specific period of time that that stratum represents. Consider the images to the right and below. On the right is a type of arthropod called a Trilobite. These animals crawled on the seafloor during the Paleozoic era hundreds of thousands of years ago. This specific type of Trilobite, however, has a much narrower existence appearing in the fossil record during the Early Cambrian and disappearing by the Middle Cambrian (existing from approximately 521 to 500 million years ago). This taxonomic order of Trilobite, known as *Redlichiida*, has a distinctive morphology which makes it easy to identify, it was short lived to other Orders of Trilobite (only existing for about 21 million years), and it was abundant and widespread during its existence. For a paleontologist, this fossil represents a specific moment in time. If one of these fossils is discovered during excavation the team can be confident that they are working in early-to-middle Cambrian beds.

While some fossils can provide fairly narrow age ranges, most biostratigraphy is achieved through *indexing* multiple fossils from the same deposit. The image below illustrates a more realistic use of Index Fossils than the trilobite example. It shows an array of 7 fossils dispersed across a *stratigraphic column*, each associated with a white arrow indicating the species' existence in the fossil record. Note that some of the age ranges overlap in some strata but not in others. This is how narrow age ranges are achieved for strata through biostratigraphy. Any one of these fossils could represent an age range of hundreds of thousands of years, but to find five different Index Fossils that overlap in a single stratigraphic horizon could narrow this age range down to tens of thousands of years.

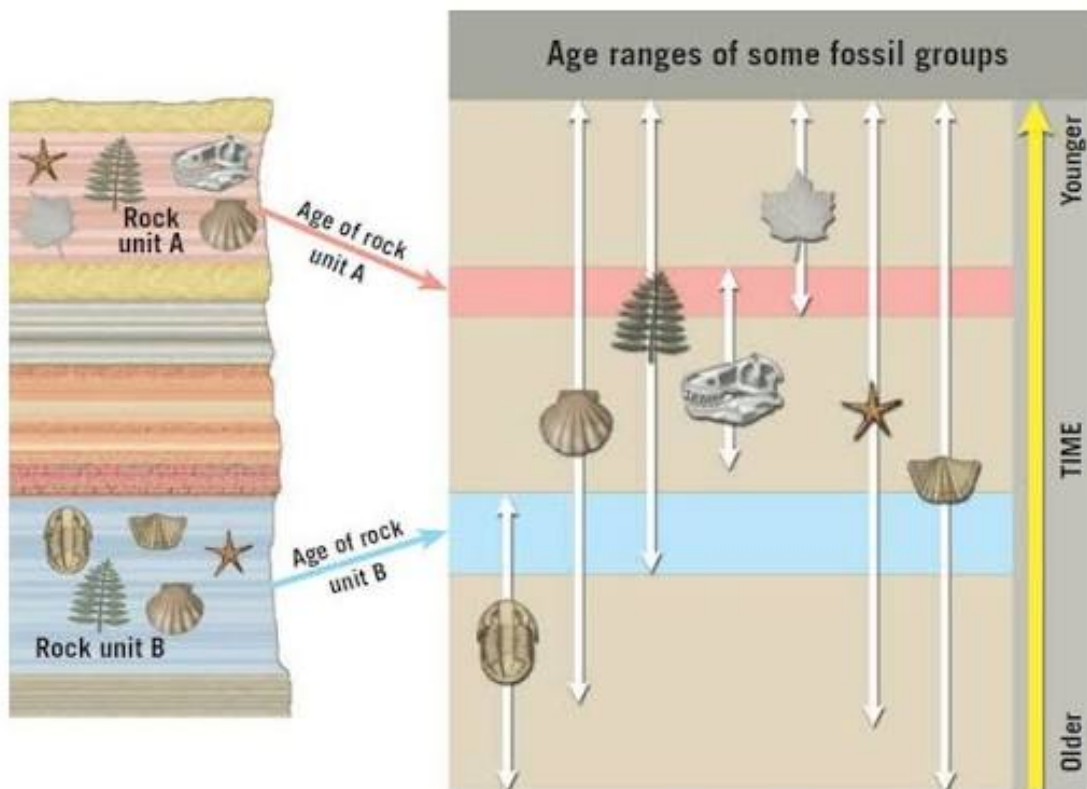


Figure 6: Using Index Fossils

The final application of Biostratigraphy is to apply it alongside of Absolute Dating techniques. We will come to learn that there are only a handful of methods that can provide a numerical age range for fossil sites and they tend to be very specific about the materials they are being used on. For example, some of these methods can only be applied to volcanic rocks while others are only able to be applied on flowstones (stalactites and stalagmites) and corals. With such narrow applications it is no surprise that there are plenty of paleontological contexts where there are no options for absolute dating at all. This is where, again, Index Fossils come into play. We already know that the right fossil in one of these sites can be correlated with a stratum elsewhere in the world. Now imagine that the correlated stratigraphy also exists in a region where the right materials for Absolute Dating exist. All of the sudden your correlated fossil can be directly associated with a numeric age.

The image below helps exemplify this process. Consider for a moment that you are a researcher in Area A. This is the only site that does not have the needed material to obtain a numeric age for an assemblage. Despite this, it is still your goal to figure out roughly how old some of these fossils are. Let's take the red spiral shell fossils, for instance. You know that similar fossils have been found in different parts of the world. In fact, Areas C, B, and D each seem to have a stratigraphic timeline similar to the progression of fossils found in your site. Knowing this, and knowing that the other research areas have absolute dates associated with certain fossils, can you come up with an estimate for the age of the red spiral shell fossils in your assemblage? You should be able to

come up with a rough age of 60 Ma. The red spiral shell fossils, when considering all research areas, are sandwiched between a 50 Ma and a 70 Ma date range. Again, not perfect, but certainly better than nothing.

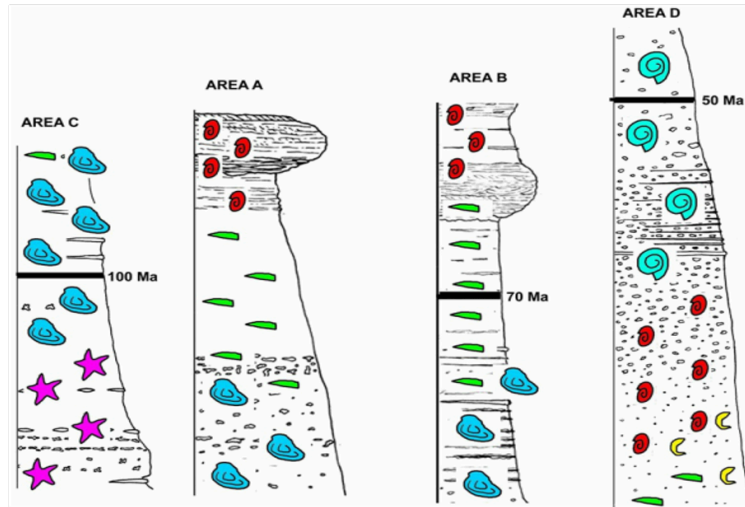


Figure 7: Biostratigraphy and Absolute Dating

4 Absolute (Radiometric) Dating



Absolute dating is better referred to as Radiometric (or Chronometric) dating. This is because, despite refined methods, seldom are dates “absolute”. This was discussed briefly above (in the opening paragraphs of the **Dating** section). These dates always come with a confidence interval (age range) dictated by a margin of error. Thus in the figure above we saw an “Absolute” date of 120 ± 5 . Here the margin of error is ± 5 and the confidence interval is 115-125 (*for those familiar with statistics you can attribute the margin of error to one or, ideally, two standard deviations*). The point is, an “Absolute Date” does not necessarily provide an exact date. This is the influence for moving away from the term “Absolute” and shifting to Radiometric. The term Radiometric is the preferred replacement for Absolute because it conveys the methodological approach to these dates. Numerical ages provided in Archaeology and Paleontology typically come from understanding the **Radioactive Decay of Isotopes** in Geologic and Organic matter (though there are a few non-Radiometric methods...hence the use of Chronometric by some camps).

Your book does well to cover the fundamentals of Radiometric Dating needed for this class. Be sure you understand the following terms/concepts:

! Important

Be sure to understand these details from the text:

- What is the primary composition of an Atom?
- What are isotopes?
- Why do some isotopes decay?

One thing that is barely mentioned which is important to understanding this process is the **Half-Life**, so let's discuss that here for a moment.

4.0.a Radioactive Decay and Half-Life

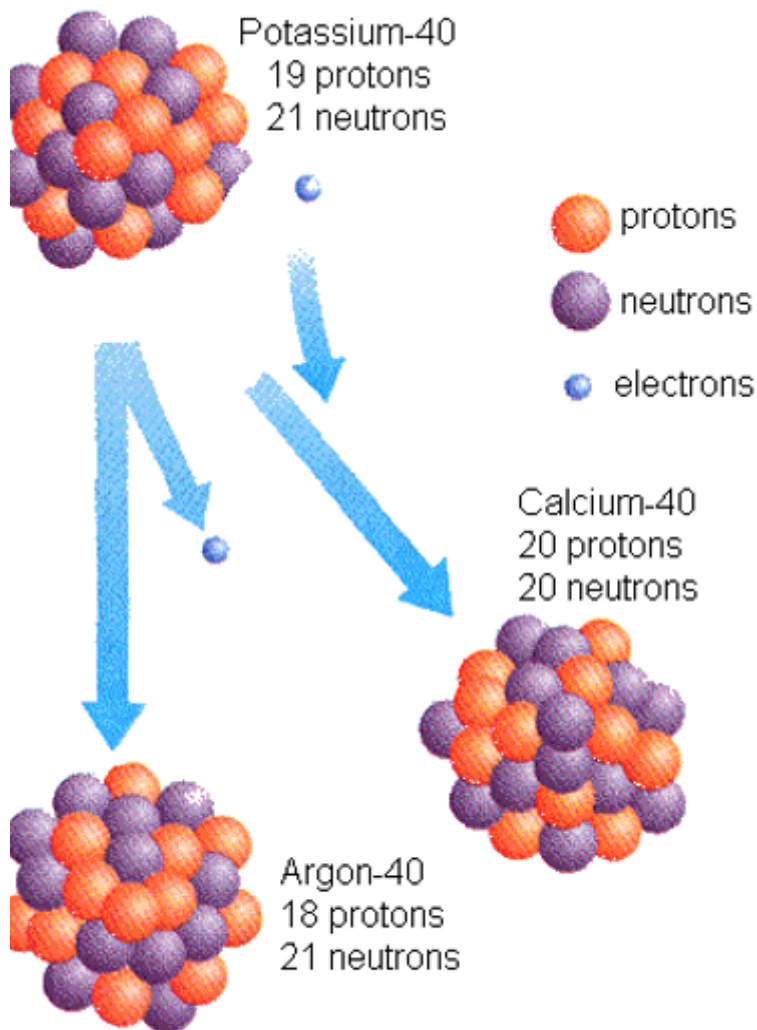


Figure 8

Hopefully by now you understand that there are unstable atoms and that they eventually stabilize by going through the process of radioactive decay. That's all well and good, but how does that help us? We use our knowledge of decay rates of different isotopes to determine how much time has elapsed since the material that we are studying first formed. As an example, when a volcano erupts it creates volcanic rocks that are enriched with radioactive isotopes. This happens because of the immense amount of energy coursing through the materials. This energy destabilizes the atomic structure of the materials, making them radioactive. The image on the right depicts this process. Volcanic ash is largely comprised of radioactive potassium (^{40}K). As the ash cools down, and the energy dissipates, these radioactive potassium nuclides spontaneously breakdown (decay) into more stable forms (^{40}Ar). The moment this process starts is the moment that the Radiometric Clock begins ticking. As long as we know how long it takes for a population of radioactive atoms

to decay into stable forms, we can judge how much time has passed since the beginning of the clock. In this example, it would provide us with the amount of time that has elapsed since the eruption of the volcano. This is what we mean by “resetting” or “zeroing” the clock. Easy, right?

4.0.a.a Understanding Half-Life

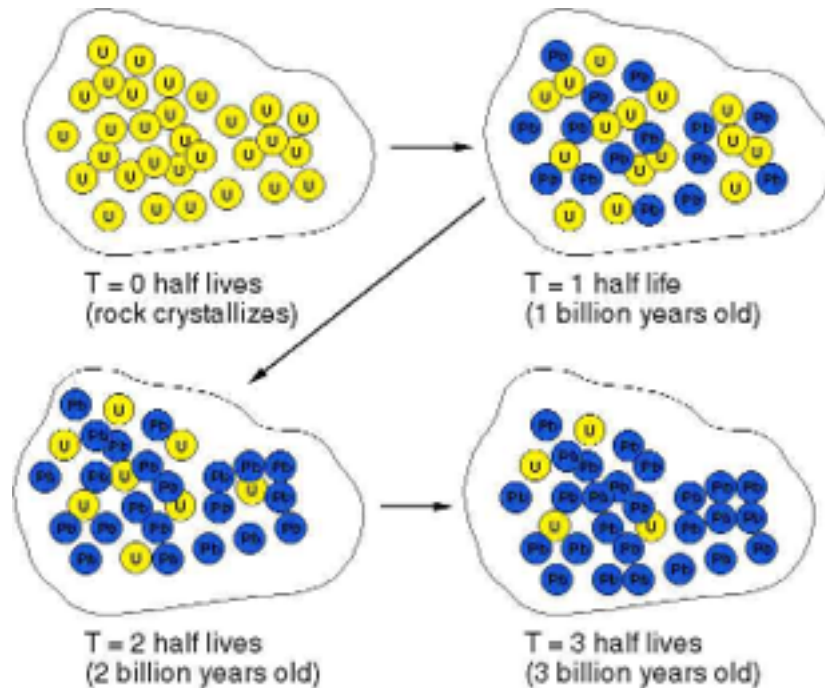


Figure 9

When we measure decay rates, we are not measuring how long it takes for one radioactive atom to decay into a stable form. Instead, we are measuring how long it takes for a population of atoms to decay from their “parent” forms (original radioactive isotope) into a population of daughter atoms. We thus investigate the changing proportion of radioactive isotopes to stable isotopes in our sample material (volcanic rock, flowstones, bones, wood, and etcetera). In the previous example we would measure the abundance of Potassium 40 to Argon 40 in volcanic rock (pumice). It is also important to understand that radioactive decay is spontaneous. As mentioned above, we can only measure the pace of decay through relative terms (not the time it takes for one atom to decay into the daughter product). This means that we are not counting how many atoms there are in a given sample. Instead, we investigate the proportion of parent to daughter atoms. Thus, our unit of measure is not an actual count of atoms. Instead we use the proportional measure of a **half-life**. This is the amount of time it takes for a sample to lose half of its radioactive material through decay. The image to the left relates the decay of radioactive Uranium to Lead, stating that the half-life is 1 billion years. Thus it takes 1 billion years for half of the radioactive uranium to decay into the daughter product of lead. After 1 billion years, therefore, there should be a 50-50 ratio of Uranium to Lead. Significantly, after another billion years, you would then observe half of that remaining product! Do see the trick? If we considered radioactive decay to be a regular and linear process, logic would dictate that after the subsequent billion year there would be no

more parent product. But that is not what this is saying. The model for radioactive decay suggests that it takes a certain amount of time for half of the radioactive product to decay. Not a specific amount (i.e., certain number of atoms). Just a proportion. But that is all we need.

A simplified example should help. Hypothetically consider that we, as humans, lose our fingers at a fixed rate over time throughout our lives. Assume the half-life for this is 40 years (meaning that it takes 40 years to lose half of our fingers). Thus, at 40 years old the average person would only have 5 fingers. But what about an 80-year-old? In a linear model of decay, we would expect the 80-year-old to have zero fingers. If you naturally lose 5 fingers over the course of forty years, another 40 years would mean another 5 fingers. This is not the case with radiometric decay. The decay rate of radioactive isotopes is one that is proportional to the larger composition. With a proportionally guided rate of decay, and a half-life of 40 years, the 80-year-old in the example would have 2.5 fingers remaining.

4.0.b Review of Carbon Dating

This is a topic also covered pretty well by the textbook but it is an important method worth reviewing here as well. So how does it all work? Carbon is a naturally occurring element that is found in all lifeforms (carbon-based lifeforms...). It has an atomic number of 6 (6 protons) and three common isotopes (12, 13 14). Carbon 14 (^{14}C) is radioactive. It originates in the atmosphere as Nitrogen 14 that is irradiated by the sun's cosmic waves. The high-energy neutrons of these cosmic rays collide with Nitrogen 14 atoms (Atomic Number=7, Mass Number=14). When these particles collide, the incoming neutrons knock out a proton thus losing a proton and gaining a neutron. This accounts for the creation of ^{14}C .

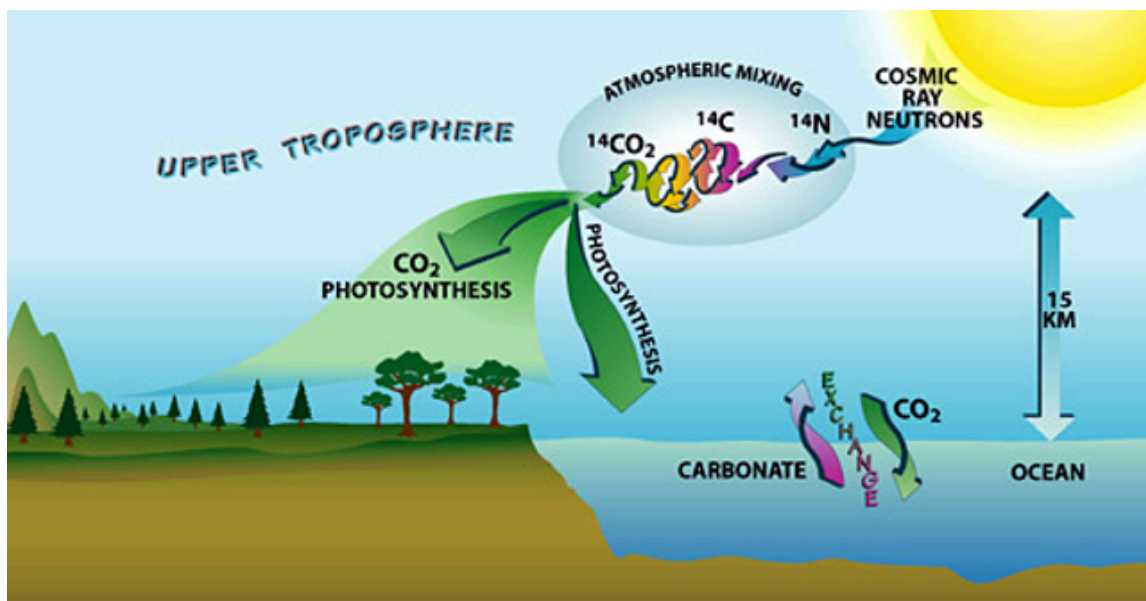


Figure 10

^{14}C is found in the same materials as all other carbon isotopes. This is because, while in the atmosphere it binds to oxygen atoms to create carbon dioxide (CO_2). Atmospheric CO_2 is then cycled into the Biosphere (through metabolic pathways) and the Hydrosphere (through dissolution). For

carbon dating we are more interested in the biosphere than the hydrosphere. This is because the materials that we are dating with this method are the organic remains of organisms that consumed carbon (which is all life on our planet). CO₂ is consumed by plant life via photosynthesis, plant life is consumed by animals, and animals are consumed by other animals. Thus, carbon is constantly cycled through the biosphere, including radioactive carbon. This doesn't mean that the carbon isn't also always decaying but as long as we continue consuming carbon, the amount of ¹⁴C in our bodies will be the same as everything else consuming carbon.

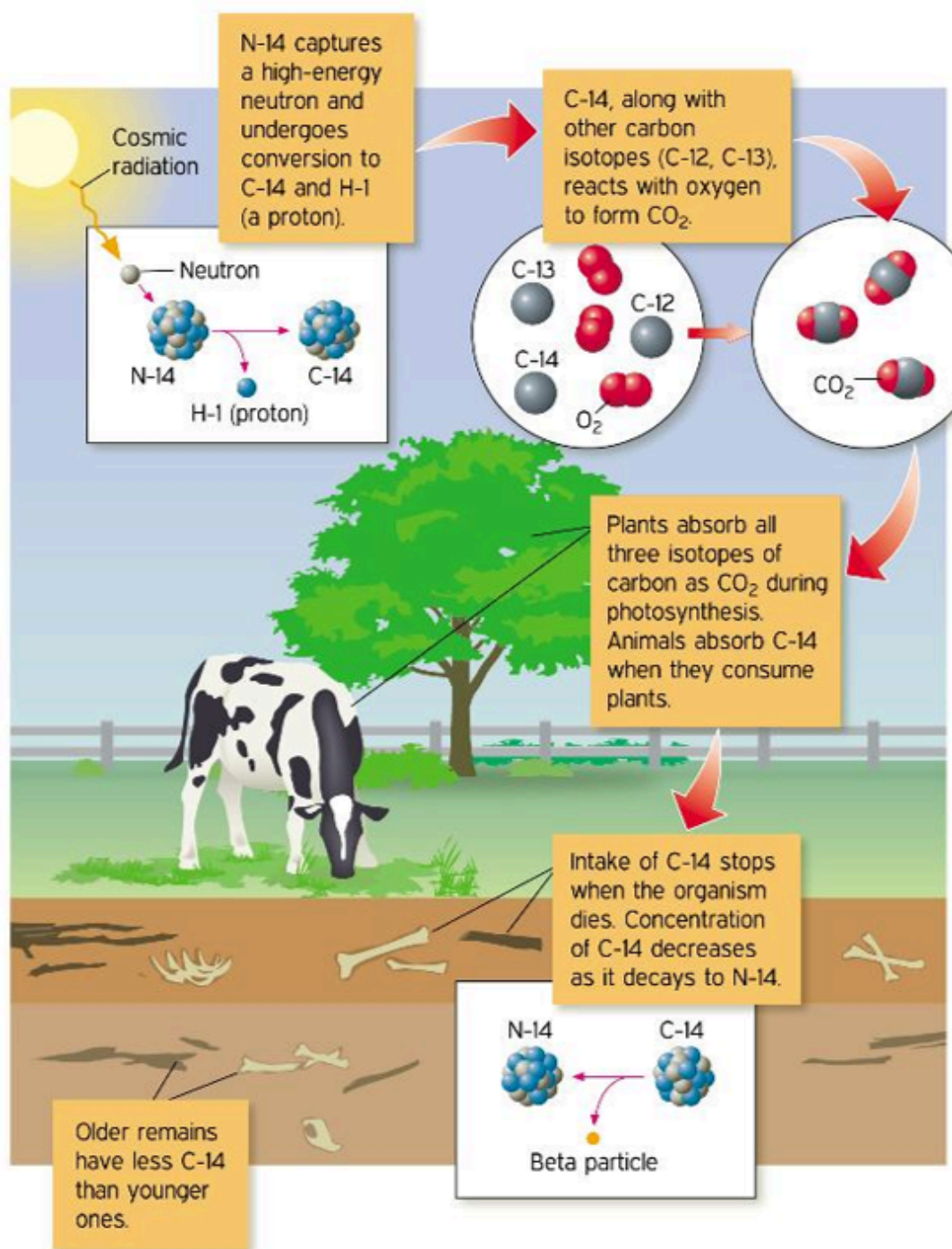


Figure 11

Regarding the radiometric clock, this starts at an organism's death (when it stops consuming ^{14}C). From here on out, the only ^{14}C activity occurring in our tissues is decay. No more uptake. The decay process is similar to the initial irradiation. Beta decay commences by ejecting an electron and changing a neutron into a proton. Thus ^{14}C (6 Protons and 8 neutrons) turns into ^{14}N (7

protons and 7 neutrons). Since ^{14}N is a gas, it is a byproduct which disperses into the atmosphere upon its inception, making it impossible for us to measure the proportion of ^{14}C to ^{14}N . But since we know how much ^{14}C is in a living structure (or in the environment), we can see how much has remains in the artifact relative to how much there would be in the “zeroed” clock. From there the process is the same as most radiometric methods. We build our timeline from the knowledge of ^{14}C having a 5730-year half-life.

Carbon dating is often considered the most ideal radiometric technique being that it is the only method that can measure radioactive decay directly from the materials we are investigating. In contrast, when we use potassium-argon dating in paleontology, we need to find specific volcanic rocks in association with the fossils we have questions about. In this way, potassium-argon dating is an indirect method (does not directly date what we want to know the age of). This is not to say that there are not limitations with Carbon Dating.

4.0.c Limitations of Carbon Dating

The most notable limitation of Carbon Dating is the short half-life of ^{14}C . Only dating materials as old as ~50-55 ka means that this method is inapplicable to many geochronological questions. As long as we stick within the field of Archaeology, however, it is mostly a non-issue. More pressing issues relate to the fluctuating concentrations of ^{14}C in our atmosphere and to the potential to confuse the method’s precision with the accuracy of its application.

4.0.c.a Calibrating the Carbon Record

Dendrochronology is the study of Tree-Ring Dating. In short, this method is grounded in the understanding that concentric rings are formed throughout a tree’s core throughout life and that the rings can be counted as successive years. Extensive records of tree-ring dates are strung together by overlapping multiple samples of trees of the same species and from the same regions. This system works because rings depicting the same patterns in different trees illustrate the same years of growth. The matching of patterns is possible because the rings vary in size and color depending on environmental conditions. Thus, the same environmental conditions in a ten-year span will create the same ring pattern on young trees and old trees of the same species in a particular area. The difference in age of these samples is important. If, say, the older tree was already 100 years old at this time, and then died shortly after, the rings for that ten-year span would be near the outer layer of the tree core (new rings form beneath the tree bark). Thus, the record is preserved at the end of a 100-year lifespan with the inner core representing the beginning of it. If we assume that the younger tree continued to grow for another 100 years after that, the same ten-year span would be found deep in the inner core of the tree. While each of these trees accounts for 100-year records of growth, overlapping the conforming ten-span provides dendrochronologists with a 200-year record. The current dendrochronological record stretches back more than 11,000 years (Friedrich et al., 2004; Leavitt and Bannister 2009).

In accounting for the known fluctuations of radiocarbon in our atmosphere, dendrochronologists work with radiocarbon labs to calibrate ^{14}C concentrations over the tree-ring record. Since carbon dating works on all organic material, ^{14}C levels can be calculated for each ring (or at least averages for sets of rings) throughout this 11,000+ year timeline. It thus detects years where the

concentrations are higher (which would mean higher baseline) or lower (lower baseline). This allows researchers to adjust their findings to the accurate year. Say, for instance, a radiocarbon analysis came back with a 50% concentration. Your instinct would be to suggest that the sample is 5,730 years old (half-life of ^{14}C). But then you match your findings with the tree-ring record and notice that that specific concentration matches with a ring that is 7,000 years old. You would then know that at this time 7,000 years ago, the ^{14}C concentrations were higher than they are today.

To use our example from earlier, years with higher concentrations of carbon would be like a person born with 12 fingers instead of 10. It would still take 40 years to lose half of their fingers, but the measurement would be of 6 fingers not 5. If you read these results not knowing that the person began with 12 fingers, you might be misled into thinking that less time has passed than actual has (29.4 years instead of 40 years).

In summary, we know that the half-life of ^{14}C is 5,730 years but that the baseline atmospheric concentration of this isotope varies through time. To use the tree-ring record, archaeologists can circumvent the issue of fluctuating ^{14}C levels by comparing the raw results of their analysis to the ^{14}C levels in dated tree-rings. While this process does sidestep the calculation of the carbon age of materials, it is important to remember that 1) the tree ring record is only approximately 11-12 thousand years old, 2) not all materials used in ^{14}C analysis are wood (bones, seeds, and etcetera), and 3) not all wood in archaeological contexts preserves the needed rings to match against the tree-ring record. Thus, beyond the evident applications of tree-ring calibration (in match specific dates), the method also informed geochronologists about the expected error in their analyses (being that carbon levels do in fact vary). This is why you will see signifiers in academic literature as it reflects Calibrated or Uncalibrated dates.

<https://www.youtube.com/watch?v=wQ7NfrN7MZU>

Good review of Carbon Dating and Calibration

4.0.c.b Precision and Accuracy

Even with the known fluctuation of ^{14}C in our atmosphere, Carbon Dating is pretty dang precise. This, however, doesn't always mean it is accurate. When using the term precision, we are simply stating that the method works. The analysis will provide you with an age for that sample that you can be confident in. But the age for that sample is not always the age for the use of that artifact. There are a couple of ways to consider the potential for inaccurate age estimations from radiocarbon analysis.

First: Probably the most ubiquitous material used in carbon dating is charcoal. This burnt wood is commonly found preserved in cooking pits from habitation areas. While the charcoal provides a great sample for carbon dating, the archaeologist must be aware that only a small section of that wood is there to sample. If the piece came from deep within the tree core, it will reflect an old age but probably not the actual age of its use in the fire. This is known as the **old wood problem**. The age of that piece of charcoal is precise but it does not accurately reflect when it was used by humans. This also means that it does not provide a reliable age estimation for the site.

Second: Archaeologists often find good samples of wood for carbon dating in ancient construction materials (say like beams used for building construction). One potential pitfall in associating the carbon date of this beam with the archaeological site is that you can't actually be sure if the tree was felled at the time of construction. The recycling of materials is not uncommon in the archaeological record. That said, when populations reuse materials salvaged from earlier settlements, archaeologists are faced with potentially misleading temporal evidence. The wooden beam in this case would date to an earlier period of time than the construction of the particular site in question.

Third: The final variation of this **old wood problem** concerns driftwood. Archaeologists here in Hawai'i know this issue well. Just as in the previous example, we have come to understand that finding wood in a site does not necessarily mean that the age of the site matches that of the wood. This is particularly the case when working in coastal environments where wood of various ages can drift on shore and be used by the populations inhabiting the area.

In all of the above cases we can see that, despite being precise, associating carbon dates with archaeological sites can still be inaccurate.